EXECUTIVE ANALYTICAL REPORT: STRATEGIC GROWTH ORIENTATION AND SYSTEMIC CREDIT RISK MITIGATION AT AURORA BANK

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I: INTRODUCTION & MANAGEMENT OBJECTIVE

Modern retail banking operations require a mathematically optimized balance between driving card-based revenue acceleration and aggressively mitigating credit default exposure. This comprehensive empirical study utilizes transactional database records, credit histories, and system infrastructure logs from Aurora Bank to deliver clear business intelligence insights.

The analytical structure segregates findings into two strategic, dedicated business dimensions:

1. **The Marketing Pillar (Branch 1):** Unlocking customer lifetime value, evaluating generational ticket sizes, and identifying premium customer acquisition targets.
2. **The Risk & Operations Pillar (Branch 2):** Isolating extreme debt-to-income (DTI) anomalies, freezing potential first-party fraud vectors, and identifying payment pipeline infrastructure bugs.

II: DATA ARCHITECTURE AND SUMMARY PERFORMANCE KIPs

To validate reporting accuracy and ensure system performance during deep drill-downs, independent decoupled tables were utilized to prevent many-to-many structural performance drops in Power BI.

The entire portfolio dataset captures a total transaction ecosystem volume of $6.87 Million spread across 157.16 thousand distinct swiping operations, maintaining an overall average purchase ticket size of $43.72.

III: PAGE 1 ANALYSIS — MARKETING & CUSTOMER VALUE ENHANCEMENT

1. Generational Revenue Contributions & Purchasing Volatility

A deep-dive review of credit card volumes across age cohorts shows that primary transaction momentum is concentrated heavily among working professionals, while a noticeable gap exists at both ends of the demographic spectrum:

- Millennials: Represent the highest revenue source at $2,511,577.03 over 54,767 transactions. This cohort yields the second-highest average transaction size at $45.86, signaling massive day-to-day card reliance.
- Generation X: Drives the highest processing velocity with 57,266 individual transactions, resulting in $2,417,111.20 in total spending. However, their average purchase amount sinks to $42.21, reflecting high-frequency, smaller-value spending habits.
- Boomers+: Contributes $1,838,916.36 across 42,999 transactions, with an average ticket of $42.77.
- Generation Z: Represents a severely under-penetrated market segment, responsible for just $106,878.90 over 2,192 transactions. Crucially, Gen Z yields the highest average ticket size at $48.76.

2. Micro-Merchant Spend Optimization

Merchant category codes (MCC) reveal that Aurora Bank cardholders heavily prioritize everyday recurring necessities and liquidity movements over luxury purchases:

- Money Transfers dominate the card pipeline at $596,560.00, indicating a heavy reliance on peer-to-peer liquidity and digital wallets.
- Grocery Stores & Supermarkets ($501,980.56) and Wholesale Clubs ($471,078.75) together comprise nearly $1M in essential volume, followed by Drug Stores/Pharmacies ($430,486.73) and Service Stations/Fuel ($380,788.89).

3. Non-User Cross-Sell Reactivation Strategy

Cross-referencing the database isolated 76 affluent, credit-worthy individuals (such as Customer ID 1325 with an income of $307,018 and Customer ID 1156 earning $280,199) who hold a perfect zero-swipe record (card_usage_count = 0). These customers are highly insulated from economic downturns but choose to leave their revolving credit limits completely untouched, representing a massive loss in potential merchant swipe fee margins for Aurora Bank.

4. VIP Campaign Selection Modeling

To answer the executive request for a customized VIP Care Program targeting the Top 10% of high-earning users with stable repayment histories, a mailing index was constructed. This algorithm identifies elite cardholders who maintain exceptional credit profiles alongside high incomes, such as Customer ID 1304 (Income: $90,812 | Credit Score: 850) and Customer ID 1223 (Income: $189,490 | Credit Score: 717).

IV: PAGE 2 ANALYSIS — CREDIT RISK & FRAUD MANAGEMENT

1. Risk Optimization: The Zero-Debt Portfolio

By plotting credit scores directly against debt-to-income (DTI) metrics, a high-value customer cluster was uncovered. This specific group of 76 customers preserves an average credit score of 756 (peaking at perfect 850 marks for Customers 21 and 1448) while maintaining a DTI ratio of exactly 0.00. This finding indicates a pristine sub-segment of consumers who have completely wiped out their active monthly liabilities, offering a zero-default footprint for immediate credit line upgrades.

2. Critical Bad Debt Alerts & Potential First-Party Fraud Identification

Conversely, the risk framework flags a group of deeply distressed borrowers who present severe default threats. A standard retail banking warning threshold triggers when a consumer's DTI surpasses 40%. The extracted alert list exposes extreme systemic anomalies:

- Customer ID 1801: Credit Score: 480 | Debt-to-Income Ratio: 2,656.96%
- Customer ID 1168: Total Debt: $242,379.00 | Credit Score: 505 | DTI Ratio: 2,652.02%
- Customer ID 1163: Total Debt: $114,251.00 | Credit Score: 489 | DTI Ratio: 2,041.11%

A DTI calculation exceeding 2,000% indicates that these cardholders owe monthly debt obligations that outweigh their regular reported income by more than twenty times. Paired with subprime credit scores crashing deep into the 400s and low 500s, these patterns suggest either severe financial distress or first-party application fraud, where an individual purposefully over-reports income parameters during onboarding to run up lines of credit with zero intent of repayment.

3. Payment Infrastructure Error Code & Gateway Analysis

An evaluation of failed transaction handshakes was structured to isolate user error from backend server dropouts across card networks:

Card Brand & Type	Insufficient Balance	Bad PIN Entries	Technical Glitch (System Error)
Mastercard Debit	**646**	**219**	**120**
Amex Credit	173	6	29
Visa Debit	*N/A*	128	29

Card Brand & Type	Insufficient Balance	Bad PIN Entries	Technical Glitch (System Error)
Mastercard Credit	145	6	29

While user behaviors like Insufficient Balance and Bad PIN numbers occur normally across all networks, Mastercard Debit processing accounts for a disproportionate 67.4% of all core technical failures (120 out of 178 total glitches). This massive concentration points to a deep software communication issue within Aurora Bank's API handshake layer or terminal integrations specific to the Mastercard Debit network.

V: STRATEGIC ACTION PLANS (ACTIONABLE INSIGHTS)

Based on the empirical findings detailed above, the following three core executive actions are recommended for immediate deployment:

1. Execute an Automated Mastercard Debit API Remediation Pipeline

- The Insight: Mastercard Debit cards dropped 120 live transactions purely due to server-side technical glitches, causing transaction friction and hurting merchant processing revenue.
- The Mandate: Task the Core IT Infrastructure Team to audit the payment gateway API handshake scripts for Mastercard debit bins within 30 days. Resolving this connection bug will reduce transaction abandonment and protect interchange fee income.

2. Deploy an Immediate Automated Credit Hold on Fraudulent DTI Accounts

- The Insight: Accounts like Customer 1168 and 1801 show extreme DTI values over 2,000% alongside failing credit scores, representing severe default risks or active fraud patterns.
- The Mandate: Risk Operations must immediately place an automated credit freeze on the top accounts flagged in the Critical Bad Debt Alert table. These files must be routed to the Fraud Investigation and Loss Recovery squads to limit financial losses before charge-offs occur.

3. Launch a Pre-Approved Cross-Selling Campaign for Zero-Debt Customers

- The Insight: A group of 76 high-earning, high-credit-score users maintain a perfect 0.00 DTI ratio, while another group of high-income individuals have a 0 card swipe history.
- The Mandate: The Product and Marketing teams should launch a combined pre-approved cross-selling campaign. Because this segment is already cleared of risk, they should be

fast-tracked for credit card activations or higher-margin products (like mortgages or auto loans) using introductory balance-transfer offers and elite travel rewards.

VI: DATA CONFIDENTIALITY & GOVERNANCE COMPLIANCE

To align this implementation with international banking standards, strict data governance policies were followed:

- **PII Redaction:** All sensitive personally identifiable information (PII)—including real names, residential addresses, and phone numbers—was scrubbed and replaced with safe, relational customer_id keys.
- **Balance Sheet Obfuscation:** Debt and income aggregates were run through numeric normalization steps (numeric_total_debt), protecting proprietary financial reporting metrics while retaining data variance for accurate modeling.